

Configuration Sensitivity in LLM Evaluation: An Artifact-Based Case Study

Shivani Bhandari

Preprint manuscript

Abstract

Large language model (LLM) benchmark scores are routinely reported as single numbers under one evaluation configuration, and model comparisons are made on that basis. We present a controlled, reproducible evaluation framework that treats the evaluation configuration—prompt template, few-shot count, inference backend, quantization, and decoding seed—as part of the measurement, and we use it to measure how sensitive benchmark scores and model rankings are to reasonable configuration changes.

Using a frozen 800-item slice (ARC-Easy, GSM8K, HellaSwag, MGSM EN/DE/FR) with generative exact-match scoring, we run a one-factor-at-a-time (OFAT) study over **three instruction-tuned models** (SmolLM2-1.7B, Qwen2.5-3B, Phi-3.5-mini) plus one single-cell Qwen2.5-7B measurement, producing **31 committed configurations** with per-item artifacts across **five factor families** (prompt, greedy seed, backend, quantization, sampling). Prompt changes dominate: within a model, prompt configuration moves overall accuracy by up to 17.4 percentage points (pp), about six times the 2.6 pp control-configuration gap between the top two models, whose Wilson intervals overlap. One legitimate prompt change (adding five exemplars) reverses the Phi-3.5-mini / Qwen2.5-3B ordering with disjoint 95% confidence intervals (Kendall $\tau_b = 0.33$ against the control ranking); quantization and backend changes move scores by at most 2.9 pp and produce no reversals. Greedy decoding is exactly seed-invariant (0 discordant items in 800), while temperature-0.7 sampling changes 9.9–24.1% of per-item outcomes with small net drift (≤ 3.5 pp).

We complement the OFAT analysis with a **post-hoc failure audit** categorizing incorrect and unparseable outputs, and we outline a planned factorial experiment (Prompt $\times$ Runtime $\times$ Seed) to analyze interaction effects that OFAT cannot recover. Paired McNemar tests with Holm correction, effect sizes (Cohen’s h), seeded bootstrap ranking stability, and an exploratory reliability composite accompany the descriptive analysis. Reported results trace to committed run manifests, configuration hashes, and raw per-item records.

We conclude that in this setting benchmark scores behave as configuration-conditional measurements, and we recommend reporting the configuration alongside interval-quantified scores. The framework, registry, and full provenance chain are released so that each number in this paper can be re-derived.

1 Introduction

Benchmark scores increasingly influence which open-weight LLMs practitioners select, which research directions are pursued, and which claims of progress are accepted. Leaderboards compress each model’s behavior on a benchmark into a single number, and model comparisons are routinely made by comparing those numbers across models.

A benchmark score, however, is not produced by the model alone. It is produced by a measurement pipeline: a prompt template and instruction phrasing, a number of few-shot exemplars,

a chat template applied to the rendered prompt, an inference backend and its version, a numerical precision or quantization scheme, a decoding strategy with its seeds, and the hardware the model runs on. Each of these is an operator choice that a reasonable evaluation could set differently without changing the scientific question being asked.

Prior work has documented that such choices matter. Prompt formatting alone can spread performance by tens of accuracy points [23], single-prompt evaluations can be brittle enough that ranking claims do not survive a multi-prompt analysis [20], and adversarial prompt perturbations reliably degrade LLM performance [29]. Evaluation-harness developers have catalogued reproducibility hazards spanning backend, precision, and prompt handling [4], and statistical treatments of evaluation as an experiment have argued for interval-based reporting [19].

Nevertheless, many published comparisons—particularly comparisons of open-weight models assembled from heterogeneous runs—still rest on a single configuration per model, and the interaction between configuration sensitivity and *model ranking* is rarely quantified directly. A score change that leaves the ordering intact supports different conclusions than one that swaps first and second place. Quantifying that distinction requires per-item paired artifacts measured across multiple configuration dimensions and models, which few evaluation setups publish alongside their scores.

This paper addresses that gap with a controlled empirical study built on a configuration-aware evaluation harness. The harness records a full manifest for every run—model and tokenizer identity, backend and version, chat template, prompt variant, decoding parameters, seed, quantization, hardware, package versions, and a SHA-256-derived configuration hash—writes per-item correctness records, and registers every completed run in a hash-deduplicated registry that binds the artifact to its configuration. On this infrastructure we run a one-factor-at-a-time (OFAT) study: one control configuration per model, with a single factor varied at a time across prompt variant, greedy seed, inference backend, quantization level, and an explicitly sampled decoding arm, on a frozen 800-item benchmark slice held fixed across all runs. OFAT isolates each dimension’s marginal effect relative to the frozen baseline; it does not estimate interaction effects between dimensions, a limitation we return to in Section 9. The measured results are: (i) prompt configuration is the dominant source of score variability, moving a model’s overall accuracy by up to 17.4 pp—about six times the 2.6 pp control-configuration gap between the two strongest models, whose control Wilson intervals overlap; (ii) a single legitimate prompt change (adding five few-shot exemplars) reverses the Phi-3.5-mini / Qwen2.5-3B ordering with disjoint 95% Wilson intervals ($\tau_b = 0.33$ against the control ranking, one pairwise reversal), while the other seven shared configurations preserve the control ordering; (iii) quantization (INT8/INT4 via bitsandbytes) and backend (Hugging Face `generate` vs. vLLM on identical rendered prompts) move scores by at most 2.9 pp and 2.4 pp, and neither change is statistically significant after Holm correction over the full 27-contrast family; (iv) under greedy decoding, changing the RNG seed is an exact no-op (0 discordant items out of 800 in every contrast), whereas temperature-0.7 sampling changes 9.9–24.1% of per-item outcomes while shifting net accuracy by at most 3.5 pp.

This paper makes the following contributions:

1. **A configuration-aware evaluation framework** that records the full evaluation context (manifest, configuration hash, git commit, hardware, package versions) for every run, documents incomparability explicitly, and registers results in a hash-deduplicated registry with per-item artifacts (Sections 2 and 12).
2. **A controlled OFAT sensitivity study** over five factor families on frozen official benchmark slices, with paired per-item statistics—Wilson intervals, exact McNemar tests with Holm correction over the 27-contrast family, paired bootstrap difference intervals, and supplemen-

tary Cohen’s h effect sizes—quantifying how much each dimension moves scores (Sections 5 and 5.3).

3. **A ranking-stability analysis** that separates score movement from ordering movement: Kendall’s τ_b against the control ranking, pairwise reversal counts, per-configuration ranks, pairwise win rates over shared configurations, and a seeded bootstrap of ranking stability—including a measured, interval-supported ranking reversal under a prompt change (Section 7).
4. **An analysis of decoding stochasticity versus determinism** quantifying both the per-item churn and the net drift induced by temperature-0.7 sampling, and the exact seed-invariance of greedy decoding, in the same harness (Section 6).
5. **A post-hoc failure audit** categorizing incorrect and unparseable outputs across all 24,800 generated outputs, with representative examples and failure category analysis (Section 5.4).
6. **A planned factorial experiment** (Prompt $\times$ Runtime $\times$ Seed) to analyze interaction effects that OFAT cannot recover, with committed configuration (Section 2.4).
7. **Reproducibility artifacts** binding every reported number to its configuration hash, run manifest, git commit, and raw per-item records, with scripts that regenerate every table and figure in this paper from those artifacts (Section 12).

We frame the measurement question formally in Section 2, describe the experimental setup in Section 4, and present results in Section 5–Section 7. Section 8 discusses implications and the distinction between score and ranking instability; the provisional reliability composite and exploratory diagnostics appear there. Section 9 states limitations explicitly, organized as threats to validity. We claim no results beyond the measured models, tasks, configurations, and hardware documented here.

2 The Evaluation Framework

2.1 Measurement formalization

Let M be a set of models, T a set of tasks, and C a space of evaluation configurations, where a configuration $c \in C$ fixes the prompt variant, few-shot count, chat template, inference backend, quantization, decoding strategy, seed, and hardware/software environment. For a benchmark slice of n items with gold answers, the measured score of model m on task t under configuration c is

$$S(m, t, c) = \frac{1}{n_t} \sum_{i \in D_t} \mathbb{I}[\hat{y}_i(m, c) = y_i], \quad (1)$$

where D_t is the frozen item set for task t ($|D_t| = 200$), $\hat{y}_i(m, c)$ is the extracted answer from the generated text, and y_i is the gold. The overall score pools the four tasks ($n = 800$). Conventional reporting emphasizes a single value $S(m, t, c_0)$ under one fixed configuration c_0 ; this work measures the set $\{S(m, t, c) : c \in C_{\text{measured}}\}$ over a controlled neighborhood of c_0 and asks what it implies for scores and rankings.

Define the pooled score $\bar{S}(m, c) = |T|^{-1} \sum_{t \in T} S(m, t, c)$; this equals item-pooled accuracy because the task slices have equal size. A model *ranking* under configuration c is $R_c(m)$, the competition rank (1 = best) by $\bar{S}(m, c)$. Equal scores receive equal ranks; subsequent ranks skip the tied

positions. Ranking stability across configurations is summarized by Kendall’s τ_b between the control ranking R_{c_0} and each variant ranking R_c [12], together with the number of pairwise reversals. We additionally report the pairwise win rate

$$P(A > B) = \frac{|\{c \in C_{\text{shared}} : \bar{S}(A, c) > \bar{S}(B, c)\}|}{|C_{\text{shared}}|}, \quad (2)$$

over configurations where both models were measured.

2.2 Interval-aware comparison

Every accuracy is reported with a Wilson score interval [27], which remains well behaved at the small n and extreme proportions that occur in our cells. Marginal interval overlap is reported descriptively, not as a test of equal model performance. Overlap neither establishes equivalence nor rules out a paired difference. Wilson intervals assume binomial sampling; the fixed heterogeneous task mixture and possible dependence between items limit this interpretation. Paired item-level comparisons use McNemar’s test [18] with continuity correction on the discordant pairs, which tests the paired hypothesis that the two configurations produce the same per-item correctness probability; because it uses pairing, it can detect changes even when unpaired intervals overlap. Effect sizes are reported as Cohen’s h [7]. Because we evaluate a family of 27 OFAT contrasts against controls, we correct for multiple testing with Holm–Bonferroni [11] and report Benjamini–Hochberg false-discovery-rate-adjusted values [3] in the appendix. Ranking-level uncertainty is quantified with a seeded bootstrap that resamples configurations (not items) and recomputes rankings.

2.3 What the framework records

Every run writes a JSON artifact containing: the full settings block (model ID, tokenizer ID, `revision`, backend, chat template, prompt variant, system prompt, `max_new_tokens`, seed, dtype, quantization, temperature, top- p , batch size, data path, and experiment ID); package versions (PyTorch, Transformers, vLLM, harness); the hardware manifest (platform, Python version, GPU model, CUDA availability); per-item records (item ID, task, gold, predicted, correct, raw generation, and per-item timing); per-task and overall accuracies with Wilson intervals; and a latency block. A 16-hex configuration hash is derived from the settings fields that affect the measurement; the registry stores one row per completed configuration, deduplicated by hash, and each row points at its artifact. Every artifact also embeds an *incomparability* statement listing the dimensions that would make two scores non-comparable. These records are what allow every number in this paper to be traced to its configuration (Section 12).

2.4 Planned nested factorial experiment

OFAT estimates simple effects at the control settings, not marginal effects averaged over other factors. The existing follow-up YAML specifies three prompts (default, concise, five-shot), three runtime bundles (HF unquantized, HF INT8, vLLM unquantized), and three greedy seeds (0, 1, 2): $3 \times 3 \times 3 = 27$ cells per model, or 81 for all three models. The runtime factor is bundled because vLLM quantization is unsupported in this harness. Prompt $\times$ backend is estimable at no quantization; Prompt $\times$ quantization is estimable within HF. Backend $\times$ quantization cannot be identified from this support pattern.

The plan in `configs/experiments/t4_factorial.yaml` uses a separate 200-item slice (100 ARC-Easy and 100 GSM8K), not the historical 800-item suite. Greedy seeds are determinism

checks, not independent stochastic replicates. Sampling would require a separately declared decoding arm. Before execution, model revisions, software, hardware and scoring must be frozen. Interaction analysis should compare paired differences-in-differences on identical item IDs, reporting uncertainty with item-level pairing and accounting for repeated responses and related items; seed duplicates must not inflate the effective sample size. No factorial inference was run in this revision: the available machine has an Apple GPU, not the required CUDA/T4 environment. These remain prospective analyses, not results.

3 Related Work

Evaluation harnesses and holistic benchmarks. The lm-evaluation-harness popularized configurable, task-level LLM evaluation [10], and its maintainers document evaluation pitfalls including prompt-format drift, backend differences, and missing error bars [4]. HELM frames evaluation as a matrix of scenarios and metrics [15], BIG-bench scales task diversity [25], and Dynabench argues for dynamic, documented benchmark iteration [13]. These systems provide configurable evaluation protocols. Our contribution is a bounded joint audit of operator choices and stored item outcomes, not a claim that existing harnesses cannot configure these dimensions.

Prompt sensitivity. Sclar et al. [23] measure performance spread under meaning-preserving prompt-format changes and find differences of up to tens of accuracy points, recommending ranges rather than single numbers. Mizrahi et al. [20] evaluate 20 LLMs across 39 tasks under diverse prompts and show that single-prompt rankings can be brittle. Zhu et al. [29] show robustness failures under adversarial prompt perturbations. Our study differs in scope—we use three deliberately distinct instruction configurations rather than large format distributions—but shares the premise that the prompt is part of the measurement, and extends the analysis to paired per-item artifacts, corrected paired tests, and explicit ranking-stability statistics on frozen slices.

Statistical treatment of evaluations. Miller [19] formalizes evaluations as experiments and recommends super-population confidence intervals for reported scores; tinyBenchmarks [17] studies how few items suffice to estimate benchmark performance. We adopt interval reporting and add paired item-level tests (McNemar) with family-wise correction, effect sizes, and a config-level bootstrap for ranking stability.

Backends and quantization. vLLM’s PagedAttention serving stack [14] and bitsandbytes INT8/INT4 quantization [8, 9] are widely used in open-weight evaluation; k-bit scaling laws quantify aggregate zero-shot accuracy trade-offs across many models. Our study measures these dimensions as factors on fixed slices with paired statistics, including the observation that our harness deliberately calls vLLM in completion mode so that both backends consume byte-identical rendered prompts.

Sampling temperature. Renze and Güven [22] find no statistically significant effect of temperature on MCQA problem solving between 0.0 and 1.0 at their scale. Our sampling arm complements this with per-item churn measurements: net drift can be small while 9.9–24.1% of items change correctness state.

Reproducibility and benchmark reliability. Biderman et al. [4] document sensitivity to evaluation setup and recommend transparent reporting of evaluation conventions. Our paired artifact audit operationalizes this recommendation, while distinguishing stored-output reproducibility from historical inference replication. LiveBench [26], first released in 2024 and revised in 2025, uses frequently updated questions and objective ground-truth scoring to limit contamination and judging bias. This addresses a different threat from configuration sensitivity: our frozen public slices support paired comparisons but do not establish freedom from contamination.

Quantization beyond the measured implementation. AWQ [16] uses activation information to protect salient weight channels in low-bit weight-only quantization and evaluates language-model and domain-specific benchmarks. Alongside k-bit scaling laws [8], this shows why precision results must be tied to a specific method and workload. We test bitsandbytes, not AWQ, and make no claim that the small corrected effects here generalize to other quantizers.

4 Experimental Setup

4.1 Models

We evaluate three instruction-tuned models under the full OFAT design—Phi-3.5-mini-instruct (3.8B parameters) [1], Qwen2.5-3B-Instruct [21], and SmolLM2-1.7B-Instruct [2]—plus Qwen2.5-7B-Instruct under a single INT4-quantized configuration (Table 1). The 7B model’s fp16, INT8, and vLLM cells exceed the 16 GB Tesla T4 budget of our compute environment, so Qwen2.5-7B appears in no ranking analysis and is reported as a single measurement. Model hub revisions were not pinned in the run manifests (`revision: null` resolves to the hub default at run time and was not recorded); this is a known limitation (Section 9).

4.2 Benchmarks

All runs use the same frozen 800-item slice: 200 items each from ARC-Easy [5], GSM8K [6], HellaSwag [28], and MGSM pooled over English, German, and French [24]. Slices were built by shuffling the official splits with a fixed seed (0 for evaluation items, 1 for few-shot exemplars) and taking the first 200; hub revisions are pinned in the committed data manifest, and each slice file carries a content SHA-256 in the artifact catalog (Table 2). Scoring is generative exact match: for multiple-choice tasks the model generates text and the first option letter is extracted; for math tasks the last number in the generated text is extracted. This protocol is deliberately *not* the log-likelihood multiple-choice protocol used by lm-eval leaderboards; our numbers are not comparable to leaderboard scores and we do not treat them as such.

4.3 Control configuration and factors

The control configuration c_0 for every model is: Hugging Face **generate** backend, the tokenizer’s own chat template applied by the harness (generation runs in completion mode on the rendered string), greedy decoding (temperature 0, top- p 1, seed 0), `max_new_tokens=256`, no quantization, the *default* prompt variant (zero-shot with task-type instructions: letter-answer for ARC/HellaSwag, step-by-step with final numeric answer for GSM8K/MGSM), batch size 1, on a NVIDIA Tesla T4 under CUDA. From this baseline we vary one factor at a time:

- **Prompt** (`prompt_id`): *concise* (minimal reply-only instructions, zero-shot) and *5-shot* (five frozen exemplars appended, instructions unchanged). Exemplars come from frozen official train/test splits outside the scored items.
- **Greedy seed**: the RNG seed is changed to 1 and 2 under temperature 0. These cells measure decoding determinism, not stochasticity.
- **Backend**: vLLM [14] (versions 0.24.0 and 0.26.0 across runs) replacing Hugging Face `generate` [4], with byte-identical rendered prompts.
- **Quantization**: INT8 and INT4 weight-only via bitsandbytes [9] on the HF path. Quantization is never crossed with vLLM in this harness.
- **Sampling**: temperature 0.7, top- p 0.95, seeds 0/1/2, measured for SmolLM2-1.7B and Qwen2.5-3B (the Phi-3.5-mini sampling cells are the three missing cells of the planned 34-cell matrix).

Every factor’s levels, the frozen control, and the per-hardware skip rules are declared in one committed YAML configuration (`configs/experiments/stability.yaml`), and each cell’s settings are frozen into its run manifest.

4.4 Provenance and completeness

The committed registry contains 31 completed cells with `status=ok` out of 34 planned Tesla-T4 cells (Table 12); the three missing cells are the Phi-3.5-mini sampling arm. Runs were produced on Colab Tesla-T4 sessions between 2026-08-20 and 2026-09-10 and are bound to 14 distinct git commits of the harness; three stale registry rows were corrected against their artifacts with the correction recorded in-row (appendix). All runs report $n = 800$ items; item-level pairing is exact (no dropped IDs in any contrast).

4.5 Statistical methods

We report Wilson 95% intervals [27] for accuracy. Main-table contrasts use exact conditional paired McNemar tests [18], with Holm–Bonferroni correction [11] over 27 same-model contrasts, including greedy seeds and sampling. The eight Qwen/Phi comparisons form a separate Holm family. Differences are in percentage points (pp), not relative percent. Their pointwise 95% intervals use a task-stratified paired item bootstrap (4,000 replicates, seed 0); these are conditional approximations, not simultaneous or cluster-robust population intervals.

The appendix additionally reports the legacy continuity-corrected McNemar analysis, Benjamini–Hochberg values [3], and Cohen’s h [7], a marginal standardized effect that does not encode pairing. Kendall’s τ_b [12], pairwise reversals, and win fractions summarize eight shared configurations. The exploratory ranking bootstrap resamples configurations (2,000 replicates); the sampling mean bootstrap resamples just three seeds. Neither should be confused with the paired item bootstrap. Related MGSM translations and GSM8K items weaken independence assumptions. All reported statistics are re-derived from stored item records (Section 12).

5 Results: Score Sensitivity

This section presents the core sensitivity results. A consolidated master table of all configurations is in Table 3, and the experimental matrix is diagrammed in Fig. 1. The full detailed analysis

Table 1: Evaluated models. “Cells” counts completed configurations with `status=ok` in the committed registry. Qwen2.5-7B was measured only under INT4 weight-only quantization: its fp16/int8/vLLM cells exceed the 16 GB Tesla T4 budget, so it is excluded from ranking analyses and appears in no ranking table.

Model	Hugging Face ID	Params (B)	Cells
Phi-3.5-mini	microsoft/Phi-3.5-mini-instruct	3.8	8
Qwen2.5-3B	Qwen/Qwen2.5-3B-Instruct	3.1	11
SmolLM2-1.7B	HuggingFaceTB/SmolLM2-1.7B-Instruct	1.7	11
Qwen2.5-7B	Qwen/Qwen2.5-7B-Instruct	7.6	1 (single cell)

Table 2: Frozen benchmark slices (generative exact-match; shuffled with seed 0, first 200 items per task; MGSM pooled over English, German, French). Hub revisions are pinned in the committed `data/official/SOURCES.md`; slice files carry content SHA-256 hashes in the artifact catalog. This generative letter/number extraction protocol is not the lm-eval loglikelihood protocol, so scores are not comparable to leaderboard numbers.

Task	Hugging Face dataset	Split	n	License	Hub revision
ARC-Easy	allenai/ai2_arc	test	200	CC-BY-SA-4.0	210d026f
GSM8K	openai/gsm8k	test	200	MIT	740312ad
HellaSwag	Rowan/hellaswag	validation	200	MIT	218ec52e
MGSM (EN/DE/FR)	juletxara/mgsm	test	200	MIT	b2f13d42

including per-task breakdowns and statistical tests appears in the following subsections, supported by Tables 4 and 5.

5.1 Control configuration

Table 4 reports Phi-3.5-mini at 67.000%, Qwen2.5-3B at 64.375%, and SmolLM2-1.7B at 39.750%. The paired Qwen-minus-Phi difference is -2.625 pp (95% interval $[-5.625, 0.375]$, Holm-adjusted $p = 0.470$). This does not establish equivalence. Overlapping marginal Wilson intervals are descriptive, not a replacement for the paired comparison in Table 10.

5.2 Prompt changes are the dominant factor

Fig. 2 and Table 5 give the central sensitivity result. The *concise* prompt is harmful everywhere, and its harm is concentrated in generative math: SmolLM2-1.7B drops 16.50 pp (39.75% $\rightarrow$ 23.25%), Qwen2.5-3B drops 13.13 pp (64.38% $\rightarrow$ 51.25%), and Phi-3.5-mini drops 8.13 pp (67.00% $\rightarrow$ 58.88%). All three McNemar contrasts remain significant after Holm correction ($p < 0.001$ adjusted), with small-to-medium effect sizes ($h = -0.36, -0.27, -0.17$).

Task-level breakdowns locate the damage: Qwen2.5-3B GSM8K falls from 85/200 to 30/200 and MGSM from 84/200 to 28/200 under *concise*, while ARC is nearly unchanged (192/200 $\rightarrow$ 195/200)—the minimal instruction removes the step-by-step scaffold that the math tasks depend on.

The *5-shot* prompt has model-dependent, sign-changing effects. It *helps* Qwen2.5-3B (+4.25 pp to 68.63%; raw McNemar $p = 0.0035$, not significant after Holm at 0.078), *hurts* SmolLM2-1.7B (-5.50 pp to 34.25%; Holm-adjusted $p = 0.037$), and *hurts* Phi-3.5-mini by 10.63 pp (to 56.37%; Holm-adjusted $p < 0.001$). The per-task pattern explains the Phi drop: five-shot exemplars lift its GSM8K (95/200 $\rightarrow$ 140/200) and MGSM (84/200 $\rightarrow$ 105/200) but collapse its ARC accuracy from

Table 3: All 31 archived cells and three missing Phi sampling cells. Each completed cell has 800 items. Accuracy and Wilson intervals are percentages; differences are percentage points against the same-model control. Exact McNemar p-values use Holm correction over 27 contrasts. Dashes indicate no contrast (control or the single 7B cell); no 7B control exists. Intervals are pointwise conditional approximations.

Model	Configuration	Acc. (%)	Wilson 95%	Δ (pp)	Holm p
SmolLM2-1.7B	control	39.750	[36.42, 43.18]	—	—
Qwen2.5-3B	control	64.375	[61.00, 67.62]	—	—
Phi-3.5-mini	control	67.000	[63.67, 70.17]	—	—
SmolLM2-1.7B	concise	23.250	[20.45, 26.30]	-16.500	< 0.001
SmolLM2-1.7B	5shot	34.250	[31.04, 37.61]	-5.500	0.0356
Qwen2.5-7B	int4	67.875	[64.56, 71.02]	—	—
Qwen2.5-3B	concise	51.250	[47.79, 54.70]	-13.125	< 0.001
Qwen2.5-3B	5shot	68.625	[65.33, 71.74]	+4.250	0.0709
Phi-3.5-mini	concise	58.875	[55.43, 62.23]	-8.125	< 0.001
Phi-3.5-mini	5shot	56.375	[52.92, 59.77]	-10.625	< 0.001
SmolLM2-1.7B	int8	41.750	[38.38, 45.20]	+2.000	1
SmolLM2-1.7B	int4	38.625	[35.31, 42.05]	-1.125	1
SmolLM2-1.7B	vllm	42.000	[38.63, 45.45]	+2.250	0.0646
Qwen2.5-3B	vllm	66.750	[63.41, 69.93]	+2.375	0.726
SmolLM2-1.7B	greedy seed 1	39.750	[36.42, 43.18]	+0.000	1
SmolLM2-1.7B	greedy seed 2	39.750	[36.42, 43.18]	+0.000	1
Phi-3.5-mini	vllm	67.125	[63.79, 70.29]	+0.125	1
Qwen2.5-3B	greedy seed 1	64.375	[61.00, 67.62]	+0.000	1
Qwen2.5-3B	greedy seed 2	64.375	[61.00, 67.62]	+0.000	1
Qwen2.5-3B	int8	64.750	[61.38, 67.98]	+0.375	1
Qwen2.5-3B	int4	65.625	[62.27, 68.83]	+1.250	1
Phi-3.5-mini	greedy seed 1	67.000	[63.67, 70.17]	+0.000	1
Phi-3.5-mini	greedy seed 2	67.000	[63.67, 70.17]	+0.000	1
Phi-3.5-mini	int8	67.250	[63.92, 70.41]	+0.250	1
Phi-3.5-mini	int4	69.875	[66.61, 72.95]	+2.875	0.726
SmolLM2-1.7B	t0.7_seed0	36.250	[32.99, 39.64]	-3.500	0.832
SmolLM2-1.7B	t0.7_seed1	37.625	[34.33, 41.03]	-2.125	1
SmolLM2-1.7B	t0.7_seed2	39.000	[35.68, 42.42]	-0.750	1
Qwen2.5-3B	t0.7_seed0	64.250	[60.87, 67.50]	-0.125	1
Qwen2.5-3B	t0.7_seed1	65.000	[61.63, 68.23]	+0.625	1
Qwen2.5-3B	t0.7_seed2	63.000	[59.60, 66.28]	-1.375	1
Phi-3.5-mini	sampling seeds 0/1/2	3 planned cells: not run			

Table 4: Control extracted-answer accuracy. Wilson intervals are conditional binomial approximations, not equivalence tests.

Model	Correct/items	Accuracy (%)	95% interval
SmolLM2-1.7B	318/800	39.750	[36.42, 43.18]
Qwen2.5-3B	515/800	64.375	[61.00, 67.62]
Phi-3.5-mini	536/800	67.000	[63.67, 70.17]

Observed OFAT coverage: 31 completed / 34 planned						
	Control	Prompt	Runtime	Seed	Sample	Total
SmolLM2	1	2	3	2	3	11
Qwen 3B	1	2	3	2	3	11
Phi mini	1	2	3	2	0/3	8/11
Qwen 7B	0	0	1	0	0	1
Runtime groups vLLM, HF INT8 and HF INT4 variants.						

Planned nested factorial: not run						
3 prompts $\times$ 3 runtime bundles $\times$ 3 greedy seeds						
Runtime: HF unquantized / HF INT8 / vLLM unquantized						
27 cells per model; 81 across three models						
Separate 200-item, two-task slice; no sampling arm						

Figure 1: Observed coverage and planned follow-up. Counts denote run cells, not independent questions. The three absent OFAT cells are Phi sampling runs. The dashed block is a separate prospective experiment, not a grid that can be completed using the historical artifacts.

197/200 to 68/200, apparently disrupting letter-answer extraction. The same factor therefore moves different models in different directions—an interaction- shaped pattern that the OFAT design surfaces qualitatively but cannot estimate formally (Section 2.4).

5.3 Quantization, backend, and sampling move scores little

Weight-only quantization changes scores by at most 2.87 pp (Phi-3.5-mini INT4: 67.00% to 69.88%, raw $p = 0.037$; not significant after Holm, $p = 0.736$). No quantization contrast survives family-wise correction. Switching to vLLM on identical prompts moves SmolLM2-1.7B by +2.25 pp (raw $p = 0.0036$; Holm 0.078) and Qwen2.5-3B by +2.37 pp (raw $p = 0.038$), with Phi-3.5-mini unchanged (+0.13 pp, $p = 1.0$). The direction is consistent (vLLM $\geq$ HF in all three cases) but no backend contrast is significant after Holm correction.

Temperature-0.7 sampling (top- p 0.95, three seeds) shifts net accuracy by at most 3.5 pp relative to greedy (SmolLM2-1.7B: 39.75% greedy vs. 36.25–39.00% sampled; Qwen2.5-3B: 64.38% greedy vs. 63.00–65.00% sampled), while per-item disagreement against the control ranges from 9.9% to 24.1% across the six sampled contrasts. Sampling noise is therefore a high-churn, low-drift process whose net effect partially cancels. Under greedy decoding, changing the RNG seed is an exact no-op: all six greedy-seed contrasts are item-for-item identical to the control (0 discordant pairs out of 800 each).

5.4 Failure analysis

Across all 31 successful runs, including sampling and the single 7B cell, 24,800 output records contain 3 unparseable outputs and no empty outputs (Table 6). These are repeated responses to 800 items, not 24,800 independent questions. An unparseable response is nonempty with a null stored prediction; a parseable wrong prediction is a separate category. Runtime failures outside successful archives are unknown.

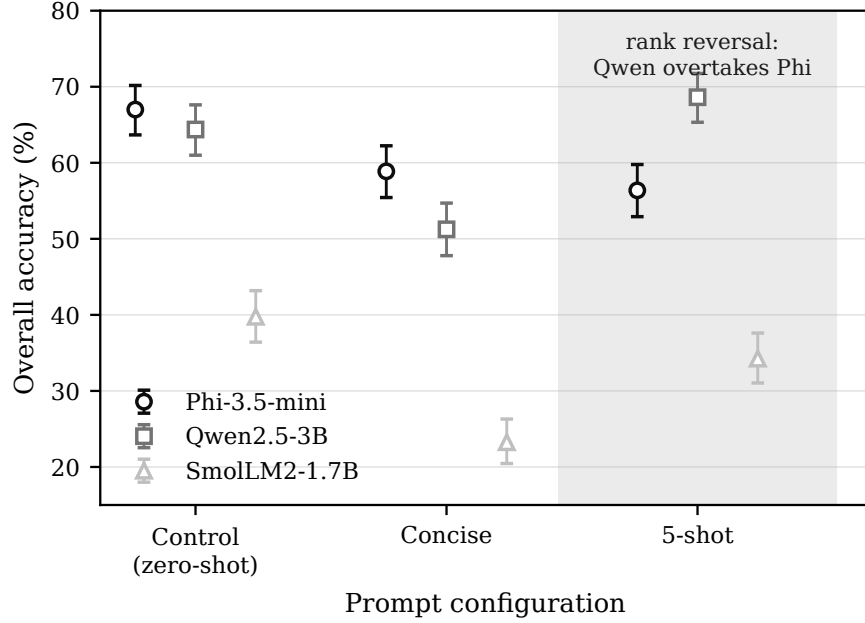


Figure 2: Prompt-dependent extracted-answer accuracy with Wilson 95% intervals.

Table 5: Variant minus control (percentage points). Task-stratified paired bootstrap intervals; exact McNemar p-values adjusted with Holm over all non-control same-model contrasts, including seeds and sampling. Intervals are pointwise, not simultaneous.

Model	Variant	Difference (pp)	95% interval	Holm p
SmolLM2-1.7B	vllm	+2.250	[0.88, 3.62]	0.0646
SmolLM2-1.7B	5shot	-5.500	[-8.88, -2.25]	0.0356
SmolLM2-1.7B	concise	-16.500	[-20.25, -12.75]	< 0.001
SmolLM2-1.7B	int4	-1.125	[-4.50, 2.25]	1
SmolLM2-1.7B	int8	+2.000	[-0.38, 4.12]	1
Qwen2.5-3B	vllm	+2.375	[0.25, 4.50]	0.726
Qwen2.5-3B	5shot	+4.250	[1.50, 7.00]	0.0709
Qwen2.5-3B	concise	-13.125	[-15.88, -10.38]	< 0.001
Qwen2.5-3B	int4	+1.250	[-1.25, 3.62]	1
Qwen2.5-3B	int8	+0.375	[-2.00, 2.62]	1
Phi-3.5-mini	vllm	+0.125	[-1.50, 1.75]	1
Phi-3.5-mini	5shot	-10.625	[-14.00, -7.38]	< 0.001
Phi-3.5-mini	concise	-8.125	[-11.25, -4.88]	< 0.001
Phi-3.5-mini	int4	+2.875	[0.25, 5.50]	0.726
Phi-3.5-mini	int8	+0.250	[-2.00, 2.38]	1

Table 6: Post-hoc output categories across all successful paper runs, including sampled and single-cell runs. Denominator counts repeated outputs, not independent questions. Runtime failures/timeouts outside archived successful runs are unknown, not zero.

Model	Outputs	Correct	Incorrect	Unparseable	Empty
SmolLM2-1.7B	8800	3296	5502	2	0
Qwen2.5-3B	8800	5619	3181	0	0
Qwen2.5-7B	800	543	257	0	0
Phi-3.5-mini	6400	4164	2235	1	0

Table 7: Selected archived failure examples (not an adjudicated prevalence estimate). Hashes identify run artifacts; IDs identify the scored items. Excerpts are verbatim prefixes or explicitly described summaries.

Category	Archived evidence and interpretation
Numeric-format failure	SmolLM2 concise, hash <code>a0852ca6fc3e5c08</code> , <code>mgsm/eval/de_0160</code> . The response begins “Am Mittwoch beträgt die Wassertiefe des Tanks zwei Drittel dessen”. It expresses a fraction in words rather than the required numeric answer. Gold: 16; prediction: null.
Refusal	Phi INT8, hash <code>cfd71ce6317d4a54</code> , <code>hellaswag/eval/7001</code> . The response begins “None of the options provided are plausible or safe.” Stored refusal flag: true; gold: A; prediction: null. This is a refusal, not evidence of a reasoning error.
Extraction mismatch	Phi five-shot, hash <code>e8c0aa94458abbd2</code> , <code>arc_easy/eval/MCAS_2001_5_19</code> . Generation begins “Answer: C” and explains photosynthesis; gold: C, stored prediction: A. This directly shows that a parseable stored answer can disagree with an explicit correct answer in the generation.
Off-target response	SmolLM2 sampling seed 1, hash <code>b3a50eddee120ab8</code> , <code>mgsm/eval/de_0038</code> . The response begins “Die Antwort betrifft die laufzeit und nicht die laufgeschwindigkeit.” It supplies no numeric answer; gold: 10, prediction: null.

Table 7 provides selected, traceable examples of numeric-format failure, refusal, off-target response, and extraction mismatch. These examples do not estimate the prevalence of reasoning errors or demonstration copying. In particular, the archived Phi five-shot response explicitly answers C while the stored prediction is A and gold is C. Rare null predictions therefore do not establish extractor validity. The prompt-level task changes are consistent with sensitivity to answer format, but a causal attribution of every error requires blinded review.

The single Qwen2.5-7B cell (INT4) scores 67.88% (543/800; Wilson [64.6, 71.0]). With no fp16 control for the 7B model we cannot attribute anything about quantization at 7B, and we do not rank this cell against other models.

6 Decoding Stochasticity versus Determinism

Two very different phenomena hide behind the word “seed.” We measure both in the same harness (Table 8, Fig. 3). For the consolidated view of all configurations including these results, see Table 3.

Table 8: Decoding stochasticity versus determinism. Sampled columns use temperature 0.7, top- p 0.95 with three different seeds; the range across seeds is in percentage points. Greedy-seed rows change only the RNG seed under temperature 0: the committed runs are item-for-item identical to the control (0 discordant pairs out of 800), so greedy seed variation contributes exactly zero measured variance in this harness.

Model	Greedy (T=0)	T=0.7, s0	T=0.7, s1	T=0.7, s2	Range (pp)	Greedy-seed no-op
Qwen2.5-3B	64.4	64.2	65.0	63.0	2.00	yes (0/800)
SmolLM2-1.7B	39.8	36.2	37.6	39.0	2.75	yes (0/800)

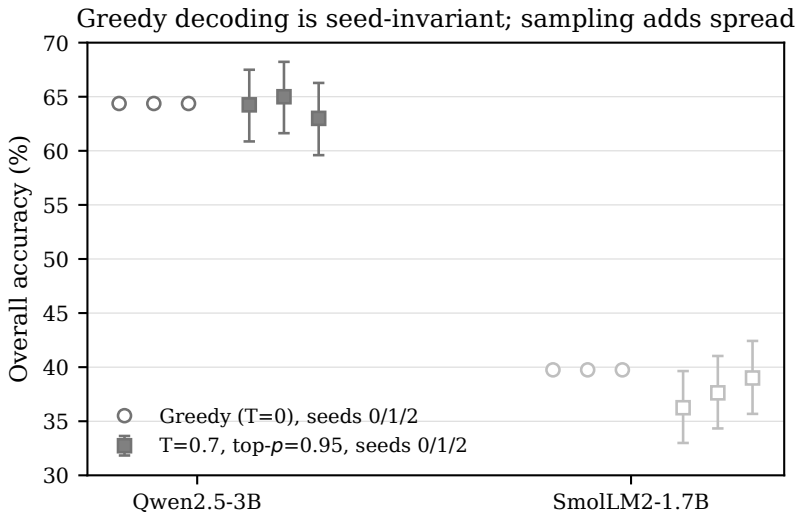


Figure 3: Greedy decoding is seed-invariant (three identical points per model; 0 discordant items in 800), while temperature-0.7 sampling (three seeds, Wilson 95% intervals) produces seed-to-seed ranges of 2.0–2.8 pp on top of a systematic drift relative to greedy.

Greedy seed changes are exact no-ops. Under temperature 0, changing the RNG seed (0→1, 0→2) reproduces the control run item-for-item in all six greedy-seed contrasts (0 discordant pairs out of 800 in each). In this harness, greedy decoding is deterministic across seeds, so greedy seed variation contributes exactly zero measured variance. Evaluations that report “different-seed greedy runs” as independent replicates should not expect them to vary under deterministic decoding; stochasticity must be introduced through sampling.

Temperature-0.7 sampling produces churn with small drift. At temperature 0.7, top- p 0.95, the three seed replicates score: Qwen2.5-3B 64.25%, 65.00%, 63.00% (range 2.00 pp); SmolLM2-1.7B 36.25%, 37.63%, 39.00% (range 2.75 pp). The seed-to-seed spread is thus comparable to the quantization and backend effects, and small compared to prompt effects. The per-item picture is very different: 9.9–24.1% of items change correctness state between any sampled run and the greedy control (the Qwen range is 9.9–11.1%; the SmolLM2 range is 21.0–24.1%). Sampling noise is therefore far from a uniform small perturbation at the item level; it is a high-churn, low-drift process whose net effect partially cancels. A single sampled run can easily land ± 2 pp away from another seed’s score purely by chance; on a leaderboard where the top-two gap is of this size (here 2.62 pp with overlapping intervals), one sampling seed is not an adequate summary of a model.

Table 9: Ranks ordered Smol/Qwen/Phi, Kendall tau-b against control, and pairwise reversals. Ties would receive shared ranks; none occur in these columns.

Configuration	Ranks	τ_b	Reversals
control	3/2/1	1.000	0
concise	3/2/1	1.000	0
5shot	3/1/2	0.333	1
vllm	3/2/1	1.000	0
seed 1	3/2/1	1.000	0
seed 2	3/2/1	1.000	0
int8	3/2/1	1.000	0
int4	3/2/1	1.000	0

Table 10: Qwen minus Phi: paired model comparison. Holm correction covers the separate family of shared-configuration comparisons. Positive values favor Qwen. Conditional, post-hoc analysis; no claim of equivalence from non-significance.

Configuration	Difference (pp)	95% interval	Holm p
control	-2.625	[-5.62, 0.38]	0.47
concise	-7.625	[-10.62, -4.62]	< 0.001
5shot	+12.250	[9.00, 15.50]	< 0.001
vllm	-0.375	[-3.38, 2.50]	0.87
seed 1	-2.625	[-5.62, 0.38]	0.47
seed 2	-2.625	[-5.62, 0.38]	0.47
int8	-2.500	[-5.50, 0.62]	0.47
int4	-4.250	[-7.38, -1.25]	0.0559

7 Ranking Stability

7.1 Scores move; orderings mostly hold

Table 9 arranges the eight configurations measured for all three fully-evaluated models and computes, per configuration, the competition ranks and Kendall’s τ_b against the control-config ranking. Under every measured factor except the five-shot prompt, the ranking is identical to the control ($\tau_b = 1.0$, zero reversals): Phi-3.5-mini first, Qwen2.5-3B second, SmolLM2-1.7B third. The five-shot prompt produces the single measured ranking change: $\tau_b = 0.33$ with one pairwise reversal, as Qwen2.5-3B (68.63%) overtakes Phi-3.5-mini (56.37%).

7.2 The reversal is interval-supported, and the control ordering is a tie

Table 10 examines the Phi/Qwen head-to-head. Under control (and under seeds, backend, and quantization) Phi-3.5-mini leads Qwen2.5-3B by +0.38 to +4.25pp, but in all of those configurations the two Wilson intervals overlap: the control ordering is a point-estimate ordering on top of a statistical tie. Under the five-shot prompt the difference is -12.26 pp *with disjoint intervals*—the only configuration in which either model statistically separates the other. We therefore state the finding conservatively: the measured configuration changes preserve a non-separated $\text{Phi} \geq \text{Qwen}$ ordering, while one legitimate prompt change produces a statistically separated ordering in the opposite direction.

Pairwise win rates over the eight shared configurations quantify the same asymmetry (Fig. 5):

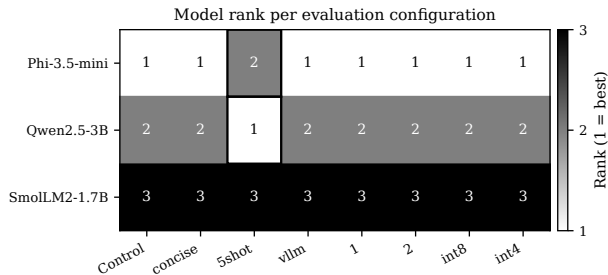


Figure 4: Model rank (1 = best) per evaluation configuration. The outlined five-shot column is the only ranking change: Qwen2.5-3B and Phi-3.5-mini swap first and second place ($\tau_b = 0.33$ vs. the control ranking, one pairwise reversal); SmolLM2-1.7B is third throughout.

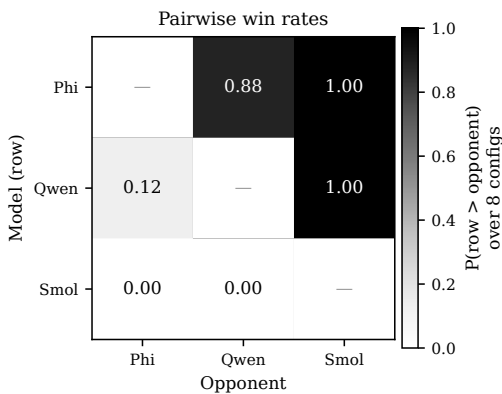


Figure 5: Pairwise win rates over the eight shared configurations: fraction of configurations in which the row model’s point estimate exceeds the column model’s. Phi-3.5-mini leads Qwen2.5-3B in 7 of 8 configurations; both lead SmolLM2-1.7B in all 8.

Phi-3.5-mini beats Qwen2.5-3B in 87.5% of configurations (7 of 8), Qwen2.5-3B and Phi-3.5-mini each beat SmolLM2-1.7B in 100%. Mean competition ranks across the eight configurations are 1.13 (Phi), 1.88 (Qwen), 3.00 (Smol); $P(\text{first place})$ is 87.5% / 12.5% / 0%. The seeded bootstrap over configurations (2,000 replicates, resampling the eight configurations with replacement) yields mean $\tau_b = 0.84$ against the mean-accuracy reference ranking and a 24.0% probability that a resampled matrix produces at least one pairwise reversal relative to that reference—i.e., the top-two order is stable in the point-estimate sequence but only weakly separated statistically, consistent with the interval analysis.

7.3 Score instability vs. ranking instability

These are distinct quantities and both matter. The concise prompt moves scores enormously (-8.1 to -16.5 pp) yet changes no ordering ($\tau_b = 1.0$): every model degrades, and the relative ordering survives. The five-shot prompt moves scores less dramatically for two of three models and yet changes the *decision* a leaderboard user would make, because the two strongest models swap places. A small score change is consequential when it crosses a ranking boundary between nearly-equal models; a large score change can be inconsequential for selection when it is uniform. This is the central practical lesson of the study for this model pair and task suite.

8 Discussion

Benchmark scores as configuration-conditional measurements. The measured pattern—large prompt effects, small backend/quantization effects, a prompt-induced rank reversal between nearly-equal models, exact greedy seed-invariance, and high-churn sampling—supports treating a reported score as a measurement produced by a model *under* an evaluation configuration, rather than as an intrinsic scalar property of the model. We state this as an empirical claim about the measured setting (three small instruction-tuned models, four generative tasks on frozen 200-item slices, one GPU family), not as a universal law; the framework exists precisely so that others can measure it elsewhere.

What follows for reporting practice. Three cheap changes to evaluation reports would have made every claim in this paper verifiable and most misreadings impossible: (1) publish the configuration (prompt template, few-shot count, backend and version, quantization, decoding, seed) next to every number, in machine-readable form; (2) publish per-item correctness records so that paired tests are possible, not just aggregate scores; (3) report interval-quantified scores and, for model comparisons, interval-overlap or paired-test outcomes rather than point-estimate ranks alone. None of these require new inference: they require recording what the runner already knows.

Score changes vs. ranking changes. The concise-prompt result shows that a factor can move scores by 8–17 pp without changing any ordering; the five-shot result shows that a factor can flip the top-two order of models whose control scores differ by only 2.6 pp. For model selection, the ranking is the decision-relevant quantity, and it is precisely the quantity most sensitive to near-ties. Evaluation reports that compress a model to one number cannot distinguish these regimes.

Provisional reliability diagnostics. As exploratory tooling (not validated methodology), the framework computes an Evaluation Reliability Score (ERS): a weighted composite of four components—CI separation among model pairs (0.667 here), seeded bootstrap ranking stability (mean $\tau_b = 0.84$), within/between config-stability (0.604), and greedy-seed stability (1.0 when the seed cells are supplied)—yielding ERS = 0.711 (3-component) or 0.740 (4-component) on our eight-configuration matrix (appendix Table 13). We present ERS as a proposed descriptive diagnostic whose components are individually meaningful, explicitly not as a validated metric; its weights are provisional and its sensitivity to weight choice can be ablated in the framework. We also ship (but do not claim results from) infrastructure for budget-constrained evaluation scheduling, which connects to efficient-evaluation work such as tinyBenchmarks [17].

Failure analysis implications. The automated categories (Table 6) contain 3 unparseable and 11,175 parseable incorrect responses across 24,800 outputs, with no empty outputs. This does not imply that parsing is harmless: the selected extraction mismatch in Table 7 shows that a correct explicit answer may still be scored incorrectly. Counts of genuine reasoning errors, demonstration copying, and extraction errors require independent adjudication and are not inferred from the null-prediction rate.

9 Threats to Validity and Limitations

We organize threats to validity following conventional categories, adapted for an artifact-based observational study.

9.1 Construct validity

What we measure may not correspond to what we claim to measure.

- **Scoring protocol.** Generative exact-match with letter/number extraction differs from the log-likelihood multiple-choice protocol used by standard leaderboards.
- **Extractor validation.** The permissive parser has not been independently adjudicated.
- **Task coverage.** Four tasks do not represent the full space of LLM capabilities.
- **Contamination.** Benchmark contamination is unmeasured here.

9.2 Internal validity

Explanations other than the configuration change may account for observed differences.

- **Software drift.** Runs span 14 distinct git commits with varying package versions.
- **Model revisions.** Model and tokenizer hub revisions were not pinned in run manifests.
- **Precision.** Recorded as `dtype=auto`.
- **Post-hoc selection.** The Phi/Qwen comparison was selected after observing top scores.

9.3 External validity

Results may not generalize beyond the measured setting.

- **Model scale.** Three models (1.7B–3.8B) plus one 7B cell.
- **Benchmark slices.** 200-item slices are small.
- **Hardware.** All runs on NVIDIA Tesla T4.
- **Sampling coverage.** Missing Phi-3.5-mini sampling cells.
- **Configuration space.** Other dimensions not explored.

9.4 Statistical conclusion validity

Statistical inferences may be weakened by assumption violations.

- **Item dependence.** MGSM translations and GSM8K questions are not independent.
- **Multiple testing.** Holm–Bonferroni is conservative.
- **Interval interpretation.** Wilson intervals assume binomial sampling.
- **Paired test assumptions.** McNemar assumes independent discordant pairs.
- **Bootstrap scope.** Ranking resamples configurations; paired differences resample items within tasks. Neither corrects for all cross-task or translated-item dependence.

9.5 Measurement and recording limitations

Data quality and completeness constraints.

- **Missing cells.** Three Phi-3.5-mini sampling cells missing.
- **Latency placeholders.** vLLM cells store 0.0 timing.
- **Reproducibility scope.** Analysis reproduction feasible.
- **OFAT, not factorial.** Full factorial (Prompt $\times$ Runtime $\times$ Seed) planned and committed to `t4_factorial.yaml`.

10 Ethical Considerations

This study uses small, publicly licensed benchmark slices (CC-BY-SA-4.0, MIT) redistributed at reduced size with attribution and license statements in the data manifest; users of the slices should consult the originals for full terms. We did not evaluate safety behaviors, toxicity, or privacy, and draw no conclusions about them; the repository’s 6-item safety canary is a pipeline check, not a safety evaluation. Benchmark contamination is unmeasured here (Section 9), and we caution against reading absolute accuracies as capability statements. No human subjects data is involved.

11 Conclusion

We presented a configuration-aware, artifact-complete evaluation framework and used it to measure how three small instruction-tuned LLMs’ benchmark scores and rankings respond to reasonable evaluation-configuration changes on frozen slices. Prompt configuration was the dominant source of variability (up to 17.4pp within a model, versus a 2.6pp top-two control gap with overlapping intervals); one prompt change produced a statistically separated reversal of the top-two ranking; quantization and backend effects were small and not significant after multiple-testing correction; greedy decoding was exactly seed-invariant; and temperature sampling induced high per-item churn with small net drift. The practical conclusion for this setting is to treat a benchmark score as a measurement conditioned on its configuration: publish the configuration, the per-item records, and interval-aware comparisons alongside every number. The framework, registry, and full provenance chain are released so that each number in this paper can be re-derived, and so that the factorial and scaling extensions needed to broaden these conclusions can be run on the same footing.

12 Reproducibility

Overarching principle. Every number in this manuscript is traceable to a committed artifact. The paper pipeline recomputes every table and figure from the same frozen sources:

1. `paper/analysis/statistics.json`, produced by `scripts/statistical_analysis.py` from the registry and run artifacts;
2. the generated tables in `paper/tables/`, produced by `scripts/generate_tables.py`;
3. the generated figures in `paper/figures/`, produced by `scripts/generate_figures.py`.

Running `python3 scripts/reproduce_paper.py` runs the full chain (`statistical_analysis.py` → `generate_tables.py` → `generate_figures.py` → `validate_paper.py`) and regenerates all downstream paper assets from the committed experiment artifacts. A validation script (`scripts/validate_paper.py`) checks registry–artifact–table consistency, sample counts, interval arithmetic, and statistical corrections.

Configuration registry. `results/registry_paper.jsonl` holds one row per completed cell with the configuration hash, model ID, factor and level, artifact path, git commit, hardware manifest, overall accuracy and Wilson 95% CI, and UTC timestamp. Two stale registry rows carry in-row correction records documenting what changed and why; the correction is recorded in the same row rather than silently overwritten.

Run artifacts. Each `results/runs/<hash>.json` contains the full settings block (model and tokenizer IDs, `revision`, backend, chat template, prompt variant, system prompt, `max_new_tokens`, seed, dtype, quantization, temperature, top-*p*, batch size, data path, experiment ID), package versions (PyTorch, Transformers, vLLM, harness), the hardware manifest (platform, Python version, GPU model, CUDA availability), all 800 per-item records (item ID, task, gold, predicted, correct, raw generation, per-item timing), per-task and overall accuracies with Wilson intervals, and a latency block. The configuration hash is derived from the settings fields that affect the measurement; a hash re-derivation check (`python -m apertus_eval_prep reproduce --check`) recomputes each hash and accuracy directly from the artifact.

Benchmark data. All runs use the same frozen 800-item slice: 200 items each from ARC-Easy, GSM8K, HellaSwag, and MGSM (EN/DE/FR). Slices were built by shuffling the official splits with a fixed seed (0 for evaluation items, 1 for few-shot exemplars) and taking the first 200; hub revisions are pinned in the committed data manifest, and each slice file carries a content SHA-256 in the artifact catalog (Table 2). Licenses: ARC-Easy CC-BY-SA-4.0; GSM8K, HellaSwag, and MGSM MIT.

Software and hardware. Runs were produced on NVIDIA Tesla T4 sessions under CUDA. Committed package versions include PyTorch 2.11.0+cu128, Transformers 5.15.0–5.16.1, vLLM 0.24.0 and 0.26.0, and the evaluation harness at versions 0.2.0–0.4.0. Runs span 14 distinct git commits of the harness between 2026-08-20 and 2026-09-10.

Test coverage. The harness test suite (33 test modules; 205 core tests pass on Python 3.14 without GPU dependencies) covers the statistics, ranking, reliability, registry, and report modules used in this paper.

Experimental matrix. Fig. 1 distinguishes historical OFAT coverage from the unrun nested factorial. Table 6 reports automated output categories; Table 7 gives selected archived examples, not manually adjudicated category prevalence.

Compute requirements. Inference on the T4 at batch size 1 across 31 cells with 800 items each was run incrementally on available GPU sessions. A full regeneration of the paper’s numbers from scratch requires re-running that inference; the analysis, table, figure, and validation steps run on CPU and complete in minutes.

Table 11: All 27 paired OFAT contrasts against each model’s control configuration (800 items per contrast). “A-only” counts items correct under the control and wrong under the variant; “B-only” counts the reverse; “Disagr.” is the total disagreement rate. Raw (continuity-corrected McNemar), Holm-adjusted, and Benjamini–Hochberg-adjusted p -values over the full 27-test family.

Model	Factor	Level	A-only	B-only	Disagr.	p (McN)	p (Holm)	p (BH)
SmolLM2-1.7B	Backend	vLLM	8	26	0.043	0.0036	0.0778	0.0137
SmolLM2-1.7B	Prompt	5-shot	115	71	0.233	0.0016	0.0372	0.0087
SmolLM2-1.7B	Prompt	Concise	194	62	0.320	<0.001	0.0000	0.0000
SmolLM2-1.7B	Quantization	INT4	102	93	0.244	0.5667	1.0000	1.0000
SmolLM2-1.7B	Quantization	INT8	36	52	0.110	0.1098	1.0000	0.2696
SmolLM2-1.7B	Sampling	T=0.7, s0	106	78	0.230	0.0465	0.8377	0.1257
SmolLM2-1.7B	Sampling	T=0.7, s1	105	88	0.241	0.2494	1.0000	0.5411
SmolLM2-1.7B	Sampling	T=0.7, s2	87	81	0.210	0.6997	1.0000	1.0000
SmolLM2-1.7B	Greedy seed	1	0	0	0.000	1.0000	1.0000	1.0000
SmolLM2-1.7B	Greedy seed	2	0	0	0.000	1.0000	1.0000	1.0000
Qwen2.5-3B	Backend	vLLM	28	47	0.094	0.0377	0.7357	0.1130
Qwen2.5-3B	Prompt	5-shot	47	81	0.160	0.0035	0.0778	0.0137
Qwen2.5-3B	Prompt	Concise	135	30	0.206	<0.001	0.0000	0.0000
Qwen2.5-3B	Quantization	INT4	47	57	0.130	0.3775	1.0000	0.7280
Qwen2.5-3B	Quantization	INT8	45	48	0.116	0.8357	1.0000	1.0000
Qwen2.5-3B	Sampling	T=0.7, s0	45	44	0.111	1.0000	1.0000	1.0000
Qwen2.5-3B	Sampling	T=0.7, s1	42	47	0.111	0.6716	1.0000	1.0000
Qwen2.5-3B	Sampling	T=0.7, s2	45	34	0.099	0.2606	1.0000	0.5411
Qwen2.5-3B	Greedy seed	1	0	0	0.000	1.0000	1.0000	1.0000
Qwen2.5-3B	Greedy seed	2	0	0	0.000	1.0000	1.0000	1.0000
Phi-3.5-mini	Backend	vLLM	22	23	0.056	1.0000	1.0000	1.0000
Phi-3.5-mini	Prompt	5-shot	180	95	0.344	<0.001	0.0000	0.0000
Phi-3.5-mini	Prompt	Concise	119	54	0.216	<0.001	0.0000	0.0000
Phi-3.5-mini	Quantization	INT4	44	67	0.139	0.0368	0.7357	0.1130
Phi-3.5-mini	Quantization	INT8	40	42	0.102	0.9121	1.0000	1.0000
Phi-3.5-mini	Greedy seed	1	0	0	0.000	1.0000	1.0000	1.0000
Phi-3.5-mini	Greedy seed	2	0	0	0.000	1.0000	1.0000	1.0000

A Appendix

A.1 All paired OFAT contrasts

A.2 Matrix coverage

A.3 Provisional reliability score

A.4 Latency

A.5 Master results table

The consolidated results appear once in Table 3.

A.6 Failure analysis

See Tables 6 and 7 for counts and archived examples.

Table 12: Committed paper-matrix coverage per model and factor. Every cell is one completed run with a frozen 800-item slice, a full manifest (settings, package versions, hardware), and a SHA-256-derived configuration hash registered in `results/registry_paper.jsonl`.

Model	Control	Prompt	Quant.	Backend	Seed	Sampling	Total
Phi-3.5-mini	1	2	2	1	2	0	8
Qwen2.5-3B	1	2	2	1	2	3	11
SmolLM2-1.7B	1	2	2	1	2	3	11
Total: 31 of 34 planned T4 cells; 3 missing (Phi sampled, seeds 0-2). Runs span 14 git commits, 2026-08-20 to 2026-09-10.							

Table 13: Provisional Evaluation Reliability Score over the eight-configuration matrix (seeded bootstrap, 2000 replicates; mean tau 0.8403, probability of at least one pairwise reversal under config resampling 0.2395). ERS is an exploratory composite proposed by this framework, not a validated metric: its components are reported individually throughout the paper, and its weights are provisional. Higher values are intended to indicate more reliable model separation.

Component	Value	Default weight
ci_separation	0.6667	0.40
bootstrap_tau	0.8403	0.30
config_stability	0.6037	0.20
seed_stability (seed scores supplied)	1.0000	0.10
ERS (3 components)		0.7106
ERS (4 components)		0.7395

A.7 Experimental matrix diagram source

The experimental matrix (Fig. 1) is drawn directly with TikZ in the results source. Its prospective nested-factorial block follows `configs/experiments/t4_factorial.yaml`; it is not an executed extension of the historical OFAT matrix.

A.8 Audited task-level breakdown

References

- [1] Marah Abdin, Jyoti Aneja, Hany Awadalla, Ahmed Awadallah, Ammar Ahmad Awan, Nguyen Bach, Amit Bahree, Arash Bakhtiari, Jianmin Bao, Harkirat Behl, et al. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*, 2024.
- [2] Loubna Ben Allal, Anton Lozhkov, Elie Bakouch, Gabriel Martín Blázquez, Guilherme Penedo, Lewis Tunstall, Andrés Marafioti, Hynek Kydlíček, Agustín Piqueres Lajarín, Vaibhav Srivastav, Joshua Lochner, Caleb Fahlgren, Xuan-Son Nguyen, Clémentine Fourrier, Ben Burtenshaw, Hugo Larcher, Haojun Zhao, Cyril Zakka, Mathieu Morlon, Colin Raffel, Leandro von Werra, and Thomas Wolf. SmolLM2: When smol goes big – data-centric training of a small language model. *arXiv preprint arXiv:2502.02737*, 2025.
- [3] Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995.

Table 14: Runtime measured per run by the harness (mean over 800 items, batch size 1, Tesla T4). Committed vLLM cells store 0.0 placeholders for timing; following the repository cost model these are treated as UNAVAILABLE rather than read as zero milliseconds, so no HF-vs-vLLM latency comparison is claimed.

Model	Config	TTFT (ms)	E2E (ms)	Tokens/s
Phi-3.5-mini	vllm	UNAVAILABLE (backend timing placeholder)		
Phi-3.5-mini	control=control	513.1	9116.9	19.93
Qwen2.5-3B	vllm	UNAVAILABLE (backend timing placeholder)		
Qwen2.5-3B	control=control	426.8	6530.2	10.81
SmolLM2-1.7B	vllm	UNAVAILABLE (backend timing placeholder)		
SmolLM2-1.7B	control=control	245.9	4015.1	23.83

Table 15: Control task-level extracted-answer accuracy; the full task breakdown for every run is exported in run details.

Model	Task	Correct	Items	Accuracy (%)
SmolLM2-1.7B	arc_easy	144	200	72.00
SmolLM2-1.7B	gsm8k	64	200	32.00
SmolLM2-1.7B	hellaswag	66	200	33.00
SmolLM2-1.7B	mgsn	44	200	22.00
Qwen2.5-3B	arc_easy	192	200	96.00
Qwen2.5-3B	gsm8k	85	200	42.50
Qwen2.5-3B	hellaswag	154	200	77.00
Qwen2.5-3B	mgsn	84	200	42.00
Phi-3.5-mini	arc_easy	197	200	98.50
Phi-3.5-mini	gsm8k	95	200	47.50
Phi-3.5-mini	hellaswag	160	200	80.00
Phi-3.5-mini	mgsn	84	200	42.00

- [4] Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, Alham Fikri Aji, Pawan Sasanka Ammanamanchi, Sidney Black, Jordan Clive, Anthony DiPofi, Julen Etxaniz, Benjamin Fattori, Jessica Zosa Forde, Charles Foster, Jeffrey Hsu, Mimansa Jaiswal, Wilson Y. Lee, Haonan Li, Charles Lovering, et al. Lessons from the trenches on reproducible evaluation of language models. *arXiv preprint arXiv:2405.14782*, 2024.
- [5] Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. In *arXiv preprint arXiv:1803.05457*, 2018.
- [6] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [7] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, Hillsdale, NJ, 2nd edition, 1988.
- [8] Tim Dettmers and Luke Zettlemoyer. The case for 4-bit precision: k-bit inference scaling laws. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.

- [9] Tim Dettmers, Mike Lewis, Sam Shleifer, and Luke Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale. In *Advances in Neural Information Processing Systems 35 (NeurIPS)*, 2022.
- [10] Leo Gao, Jonathan Tow, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Kyle McDonell, Niklas Muennighoff, Jason Phang, Laria Reynolds, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation. Zenodo, lm-evaluation-harness v0.0.1, 2021. <https://github.com/EleutherAI/lm-evaluation-harness>.
- [11] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- [12] Maurice G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938.
- [13] Douwe Kiela, Max Bartolo, Yixin Nie, Divyansh Kaushik, Anas Geiger, Zhengxuan Wu, Bertie Vidgen, Grusha Prasad, Amandeep Singh, Persi Ringshia, Zhiyi Ma, Edoardo Ponti, Sonal Barezi, et al. Dynabench: Rethinking benchmarking in NLP. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4110–4124, 2021.
- [14] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.
- [15] Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, Eric Zelikman, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.
- [16] Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for LLM compression and acceleration. *arXiv preprint arXiv:2306.00978*, 2023. URL <https://arxiv.org/abs/2306.00978>.
- [17] Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. tinybenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [18] Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947.
- [19] Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations. *arXiv preprint arXiv:2411.00640*, 2024.
- [20] Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. State of what art? A call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12:933–949, 2024.

- [21] Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [22] Matthew Renze and Erhan Güven. The effect of sampling temperature on problem solving in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7346–7356, 2024. URL <https://arxiv.org/abs/2402.05201>.
- [23] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models’ sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [24] Freda Shi, Mirac Suzgun, Markus Freitag, Xuezhi Wang, Suraj Srivats, Soroush Vosoughi, Hyung Won Chung, Yi Tay, Sebastian Ruder, Denny Zhou, Dipanjan Das, and Jason Wei. Language models are multilingual chain-of-thought reasoners. *arXiv preprint arXiv:2210.03057*, 2022.
- [25] Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, et al. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *arXiv preprint arXiv:2206.04615*, 2022.
- [26] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, Sreemanti Dey, et al. LiveBench: A challenging, contamination-limited LLM benchmark. *arXiv preprint arXiv:2406.19314*, 2024. URL <https://arxiv.org/abs/2406.19314>.
- [27] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927.
- [28] Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. HellaSwag: Can a machine really finish your sentence? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 4791–4800, 2019.
- [29] Kaijie Zhu, Jindong Wang, Jiaheng Zhou, Zichen Wang, Hao Chen, Yidong Wang, Linyi Yang, Wei Ye, Yue Zhang, Neil Zhenqiang Gong, and Xing Xie. PromptRobust: Towards evaluating the robustness of large language models on adversarial prompts. *arXiv preprint arXiv:2306.04528*, 2023.